Sanaul Haque

Kumush_Habibullaeva_Proposal

Assignment 206

Document Details

Submission ID
trn:oid:::3117:636677837

Submission Date
Oct 9, 2026, 10:59 PM GMT+6

Download Date
Oct 9, 2026, 11:03 PM GMT+6

File Name
Kumush_Habibullaeva_Proposal.pdf

File Size
103.9 KB

6 Pages

1,795 Words

11,096 Characters

20% Overall Similarity

The combined total of all matches, including overlapping sources, for each database.

Match Groups

17 Not Cited or Quoted 16%
Matches with neither in-text citation nor quotation marks

1 Missing Quotations 0%
Matches that are still very similar to source material

3 Missing Citation 4%
Matches that have quotation marks, but no in-text citation

0 Cited and Quoted 0%
Matches with in-text citation present, but no quotation marks

Top Sources

16% Internet sources
12% Publications
16% Submitted works (Student Papers)

Match Groups

- **17** Not Cited or Quoted 16%
  Matches with neither in-text citation nor quotation marks
- **1** Missing Quotations 0%
  Matches that are still very similar to source material
- **3** Missing Citation 4%
  Matches that have quotation marks, but no in-text citation
- **0** Cited and Quoted 0%
  Matches with in-text citation present, but no quotation marks

Top Sources

16% Internet sources
12% Publications
16% Submitted works (Student Papers)

Top Sources

The sources with the highest number of matches within the submission. Overlapping sources will not be displayed.

1	Student papers	Georgia Institute of Technology Main Campus on 2026-04-06 — 2%
2	Internet	jurnal.fokgii.com — 2%
3	Publication	Ping-Kuo A. Chen, Alexa Mae F. Solis. "Driving factors and moderator in adopting ... — 2%
4	Publication	Xiaoyu Chen, Zhuohong Gong, Zhaoting Ling, Ollie Yiru Yu, Falk Schwendicke, Wal... — 2%
5	Student papers	Murdoch University on 2025-07-19 — 2%
6	Internet	www.altmetric.com — 1%
7	Internet	arxiv.org — 1%
8	Student papers	University of Liverpool on 2026-01-05 — 1%
9	Student papers	East Durham College on 2024-01-23 — <1%
10	Internet	pmc.ncbi.nlm.nih.gov — <1%

11 Internet

www.crbeer.com.hk <1%

12 Internet

www.scielo.br <1%

13 Student papers

Brigham Young University on 2026-09-21 <1%

14 Student papers

Copenhagen Business School on 2026-02-09 <1%

15 Student papers

University of Surrey on 2025-09-07 <1%

16 Internet

eprints.soton.ac.uk <1%

17 Internet

www.mdpi.com <1%

18 Internet

www.medrxiv.org <1%

19 Student papers

Universitas Muhammadiyah Yogyakarta on 2026-04-23 <1%

20 Student papers

University of Hull on 2022-05-20 <1%

IARCO RESEARCH PROPOSAL

Research topic: The Role of Behavioural Design in Transforming Oral-Health Awareness into Consistent Toothbrushing Habits in Children

Full Legal Name: Kumush Habibullaeva Hikmatullaevna

Institution: Invento the Uzbek International School

Country: Uzbekistan

Category: Junior

Class/Year: 11 (IB DP1)

Major/Subject: Behavioural Science and Oral Health

Submission date: September 20, 2026

Registered email address: kumnlhk@gmail.com

Title:

The Role of Behavioural Design in Transforming Oral-Health Awareness into Consistent Toothbrushing Habits in Children

Abstract

I designed a "Tiny Teeth, Big Habits" project (TTBG)—a one-month behavioural intervention involving more than 100 primary-school children — to develop consistent brushing habits in children. Initially, I expected the problem to be inadequate awareness of the importance of consistent brushing. However, the pre-survey showed that while 94% understood the significance of proper brushing, 59% reported brushing twice daily. These outcomes revealed a clear awareness-behaviour gap: although almost every child was well informed about proper brushing, fewer than half reported consistently remembering to brush.

This research aims to identify which behavioural design strategies reduce the awareness-behaviour gap and turn oral-health knowledge into consistent brushing habits. The investigation will involve first- and second-grade children aged 6-8 years. All groups will receive basic oral-health education, but the behavioural support provided to each group will differ. The results from pre-, post-, and follow-up surveys across groups will be used to compare the effectiveness of different behavioural strategies in increasing brushing frequency, consistency, interest, and ease of starting.

The findings of this research will guide the expansion of TTBG across Uzbekistan, helping children to form health-related habits from an early age. Moreover, similar habit-formation mechanisms may appear in other fields such as studying, diet, and environmental behaviour; therefore, understanding these mechanisms could provide insights into broader contexts.

12

Introduction

Oral diseases affect approximately 3.5 billion people worldwide, or about 45% of the population. Yet most oral conditions are largely preventable, including dental caries, one of the most prevalent. A leading preventive measure is tooth brushing. Thus, having proper toothbrushing habits is a crucial part of maintaining oral and general health [1].

17
14

Importantly, understanding the significance of a practice or intention to act does not always translate into a behaviour. Feil et al. [2] investigated the intention-behaviour gap in physical activity; their analysis showed that even among people intending to be physically active, 47.6% did not convert their intention into actual behaviour. Hence, factors beyond intention may affect whether intention turns into action.

Similarly, for oral-hygiene habits, although most children in TTBG knew the importance of tooth brushing, only about 59% reported brushing twice daily. After the

one-month intervention, daily brushing and twice-daily brushing increased by 9.9 percentage points and 19.6 percentage points, respectively. However, TTBG included various behavioural strategies simultaneously; consequently, due to methodological limitations, the different techniques were not isolated, leading to a need for new research that compares the contribution of each selected strategy to increased brushing frequency and consistency. The new research will separate behavioural interventions to compare which ones are associated with greater improvements in brushing frequency and consistency among children.

Literature review

Previous research [4] shows that habit formation, unlike conscious and deliberate action, is a complex process involving multiple changes in neural activity. Initially, behaviour can be goal-directed and linked to a specific outcome. Repeating the behaviour fosters connections and associations between the cue and the action [4]. Another study [5] empirically supported this finding, showing that over time, repeated behaviour gradually became automatic. Importantly, the time to reach 95% of asymptotic automaticity varied greatly (approximately 18-254 days) [5], meaning people develop habits at their own pace.

Loewenstein et al. [3] conducted an intervention among primary-school children, giving small incentives for eating fruits or vegetables at lunchtime. During the intervention, healthy-eating behaviour increased substantially: incentives doubled the percentage of participants who ate at least one portion of fruit or vegetables. A follow-up two months after the intervention showed that the behavioural change remained partially (21-44% above baseline) [3]. These findings suggest that short-term external reinforcement can be associated with repeated health-related behaviour in children even after it is removed.

Numerous studies have investigated the role of gamification in oral health behaviour in children. Chen et al. [6] reviewed 41 such studies; the most prevalent behavioural strategies were rewards, progress tracking, and interactive feedback. Studies reported promising effects on outcomes such as oral-health knowledge, oral-hygiene/brushing behaviour, and engagement/adherence. The review concluded that gamification can increase engagement and improve oral-hygiene behaviours [6].

Nevertheless, the Happy Teeth RCT [7] evaluated the effect of gamification on children's brushing behaviour, separating participants into intervention and control groups. While both groups received standard oral-health education, the former group also used the gamified Happy Teeth app. Both groups' oral-health knowledge increased significantly, but the intervention group's increase was significantly higher. However, brushing frequency remained unchanged [7]. This suggests that gamification can improve knowledge and engagement, but it does not guarantee increased brushing frequency.

Overall, while there is potential evidence supporting the effectiveness of gamified interventions, results are not always consistent across outcomes. Methodological limitations of existing research further complicate drawing firmer conclusions. These

include heterogeneity in the reviewed studies' intervention designs, behavioural/ gamification elements, and measured outcomes [6].

A gap remains in existing research: which selected behavioural strategies best translate awareness of the importance of toothbrushing into consistent toothbrushing behaviour, and to what extent the results persist after the intervention [6], [8]. Further research is needed to compare selected behavioural methods under comparable conditions and evaluate their effect on brushing behaviour.

Research questions

- Which selected behavioural design strategies best translate oral-health awareness into consistent toothbrushing behaviour in children?
- How does each behavioural design strategy relate to changes in process-related outcomes such as ease of remembering to brush, ease of starting to brush, and interest in brushing during and after the intervention?

Methodology

This research applies a mixed-method approach, but is primarily quantitative.

Quantitative Phase

The participants are children aged 6-8 years. Final group sizes will depend on the number of children in the selected classes, since treating children in the same environment differently can create perceived unfairness. Each class will receive a specific behavioural strategy for one month:

1. Education only; 2. Education + One-Month Challenge to brush teeth twice daily; 3. Education + Progress tracker; 4. Education + Rewards; 5. Education + Oral-health Supplies; 6. Education + Challenge + Oral-Health Supplies + Progress Tracker + Rewards.

Therefore, different classes will represent different intervention conditions. 1st and 6th groups will serve as reference groups. All groups will complete three surveys: pre-intervention, post-intervention, and follow-up surveys. Behavioural outcomes (brushing frequency and consistency) and process-related outcomes (ease of remembering to brush, ease of starting to brush, and interest in brushing) will be measured.

Qualitative phase

All three surveys will include open-ended questions to identify reasons for inconsistent brushing behaviour, factors that helped children maintain more consistent brushing behaviour, and possible factors that would make brushing more enjoyable and effortless. These findings will help interpret and complement the quantitative results.

Data analysis

Quantitative data will be assessed by comparing behavioural and process-related outcomes (percentage-point increases between pre- and post-intervention surveys, and %

above baseline in follow-up survey) across groups. Similar recurring open-ended responses will be grouped and later used to help interpret quantitative findings.

Project practicalities

The intervention will be conducted at a primary school in Uzbekistan, involving three 1st-year classes and three 2nd-year classes. The initial session will take the form of a lesson (10 minutes of education, 10 minutes of pre-survey, and 10 minutes explaining a behavioural strategy specific to the group). Post-intervention surveys will be conducted one month after intervention introduction, and follow-up surveys will be conducted 2 months after the post-surveys. Before the intervention, parents of all participants will provide consent for their child to take part.

Roadblocks and Potential Limitations

One main limitation is that most data is obtained through self-report. To reduce reliance on self-report, parents will provide brief feedback on the child's toothbrushing behaviour by the end of the intervention. This will not fully eliminate self-report; however, it reduces dependence on the child's self-report alone. The second limitation is parental/home influences that can interact with the controlled behavioural strategies, preventing the research from fully isolating the strategy's effect. Additionally, since one intervention in represented by one class, class-level differences may partially confound strategy effects. Therefore, finding should be interpreted as comparative rather than definitive causal evidence.

15

Timeline

Preparation — ~2 weeks: finalise questionnaires and intervention materials, contact a school and get an agreement, get parental consent

Week 1: conduct initial sessions

Weeks 1-4: one-month intervention

Week 5: conduct post-surveys

Months 2-3: period with no intervention

End of month 3: conduct follow-up surveys

Months 4-5: analyse quantitative and qualitative data, compare results across groups, and start writing the final research paper

References

[1] World Health Organization. (2022). Global oral health status report: Towards universal health coverage for oral health by 2030. World Health Organization, 2022. ISBN: 978-92-4-006148-4. [Online]. Available: https://www.who.int/publications/i/item/9789240061484

2

[2] K. Feil, J. Fritsch, and R. E. Rhodes, "The intention-behaviour gap in physical activity: A systematic review and meta-analysis of the action control framework," British Journal of Sports Medicine, vol. 57, no. 19, pp. 1265–1271, 2023, doi: 10.1136/bjsports-2022-106640.

[3] G. Loewenstein, J. Price, and K. Volpp, "Habit formation in children: Evidence from incentives for healthy eating," Journal of Health Economics, vol. 45, pp. 47–54, 2016, doi: 10.1016/j.jhealeco.2015.11.004.

[4] K. S. Smith and A. M. Graybiel, "Habit formation," Dialogues in Clinical Neuroscience, vol. 18, no. 1, pp. 33–43, 2016, doi: 10.31887/DCNS.2016.18.1/ksmith.

[5] P. Lally, C. H. M. van Jaarsveld, H. W. W. Potts, and J. Wardle, "How are habits formed: Modelling habit formation in the real world," European Journal of Social Psychology, vol. 40, no. 6, pp. 998–1009, 2010, doi: 10.1002/ejsp.674.

[6] J. Chen, Y. Ying, M. Pang, J. Chen, T. Kang, P. Xuan, X. Chen, and W. Zhang, "The application of gamification in children's oral health management: Systematic review," Journal of Medical Internet Research, vol. 27, Art. no. e75541, 2025, doi: 10.2196/75541.

[7] B. Tahani, S. Sadeghi Boldaji, H. Telleih, M. Ahmadi, and H. Beigy Datli, "Development and evaluation of the effectiveness of a gamified oral health promotion app-Happy Teeth-in schoolchildren: A randomized controlled trial," BMC Oral Health, vol. 26, Art. no. 213, 2026, doi: 10.1186/s12903-025-07575-x.

[8] G. Kitsaras, S. El-Yousfi, K. A. Gray-Burrows, P. Day, and Z. Marshman, "Supervised toothbrushing programmes and at home brushing behaviour: a rapid review of evidence," Evidence-Based Dentistry, vol. 27, no. 2, p. 42, 2026, doi: 10.1038/s41432-026-01218-y.